P2: Urban Heat Island Mapping using Land Surface Temperature,Built-upDensity,and GIS Analysis.

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Remote Sensing and GIS using Python

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Project Code: P2

Title: Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis.

Objective: To analyze spatial patterns of urban heat islands (UHI) by deriving Land Surface Temperature (LST) from satellite data and examining its relationship with built-up density, enabling identification of high-temperature zones and urban thermal hotspots.

Study Area:

The study area for this project is Chennai district, Tamil Nadu, India. Chennai is one of India's largest metropolitan cities, located on the southeastern coast of the Bay of Bengal.



Figure 1: Area of Interest (AOI) — Chennai district boundary, Tamil Nadu, India.

Data Used:

Satellite Data:

Landsat-8 OLI/TIRS (for LST calculation)

Required Bands (Landsat-8):

- Band 10 (Thermal Infrared – TIRS) — used for LST derivation
- Band 4 (Red) — used for NDVI calculation
- Band 5 (NIR) — used for NDVI calculation

Auxillary Data:

Chennai district boundary loaded directly from FAO/GAUL 2015 Level-2 administrative boundary dataset through Google Earth Engine



Data Sources:

- Landsat-8 Collection 2 via Google Earth Engine: https://developers.google.com/earth- engine/datasets/catalog/LANDSAT_LC08_C02_T1_L2
- FAO/GAUL District Boundary: https://developers.google.com/earth- engine/datasets/catalog/FAO_GAUL_2015_level2

Data Range:

January 1, 2023 – December 31, 2023

Reference / Validation Data:

These factors were not considered in this study. The classification of the Urban Heat Island zones relied only on Landsat-8 thermal bands data analyzed using Google Earth Engine; there was no cross verification with Google Earth Pro images or urban temperature studies.

Methodology:

This project was carried out entirely using the Google Earth Engine (GEE) Code Editor. The methodology is as follows:

1. Defining the Area of Interest (AOI)

The Chennai district boundary was loaded directly from the FAO/GAUL 2015 Level-2 administrative boundary dataset using a district name filter, eliminating the need for manual AOI drawing or shapefile upload.

2. Loading Landsat-8 Image Collection

Landsat-8 Collection 2 imagery was filtered to the AOI and date range January 1–December 31, 2023, with cloud cover restricted to less than 10%. A median composite was computed to reduce remaining noise:

```
// Load Landsat-8 Image Collection
var image = ee.ImageCollection("LANDSAT/LC08/C02/T1_L2")
  .filterBounds(aoi)
  .filterDate('2023-01-01', '2023-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 10))
  .median();
```

3. Computing NDVI

NDVI was calculated from Band 5 (NIR) and Band 4 (Red) to support emissivity estimation:

```
// Compute NDVI
var ndvi = image.normalizedDifference(['SR_B5','SR_B4']).rename('NDVI');
```

4. Extracting Land Surface Temperature (LST)

Band 10 (Thermal Infrared) was converted from digital numbers to LST in degrees Celsius using the standard Landsat-8 scaling formula:

```
// Extract thermal band and convert to LST (Celsius)
var lst = image.select('ST_B10')
  .multiply(0.00341802)
  .add(149.0)
  .subtract(273.15)
  .rename('LST');
```

5. Visualization

The LST image was displayed using a blue–yellow–red color palette, where blue represents cooler surfaces and red represents high-temperature urban zones.

```
// Visualize LST
Map.addLayer(lst, {min:20, max:45, palette:['blue','yellow','red']}, 'LST');
```

6. Identifying Urban Heat Zones

Pixels with LST exceeding 35 °C were classified as Urban Heat Island zones and visualized separately in red:

```
// Identify Urban Heat Zones (LST > 35°C)
var uhi = lst.gt(35);
Map.addLayer(uhi.updateMask(uhi), {palette:['red']}, 'UHI Zones');
```

7. Computing Area Statistics

Mean LST across the entire AOI was computed using reduceRegion(). The total area of UHI zones (LST > 35 °C) was calculated in square kilometers by summing pixel area within the masked UHI layer.

```
// Mean LST statistics
var stats = lst.reduceRegion({
  reducer: ee.Reducer.mean(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});
print('Mean LST for AOI (°C):', stats);
```

8. Export

The final LST raster was exported to Google Drive as a GeoTIFF for further use:

```
// Export
Export.image.toDrive({
  image: lst,
  description: 'LST_Chennai_Map',
  scale: 30,
  region: aoi
});
```

```
mports (1 entry)
var geometry: Geometry (Error)
GeometryConstructors.MultiGeometry: Geometry coordinates do not have the same geodesic
state. Specify an explicit geodesic state to reproject inputs.

/ Define Chennai AOI using official boundary
ar aoi = ee.FeatureCollection("FAO/GAUL/2015/level2")
  .filter(ee.Filter.eq('ADM2_NAME', 'Chennai'))
  .geometry();
ap.centerObject(aoi, 10);
ap.addLayer(aoi, {color: 'red'}, 'Chennai District');
rint('Total AOI area (sq km):', aoi.area(1).divide(1e6));

/ Load Landsat-8 Image Collection
ar image = ee.ImageCollection("LANDSAT/LC08/C02/T1_L2")
  .filterBounds(aoi)
  .filterDate('2023-01-01', '2023-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 10))
  .median();

/ Compute NDVI
ar ndvi = image.normalizedDifference(['SR_B5','SR_B4']).rename('NDVI');

/ Extract thermal band and convert to LST (Celsius)
ar lst = image.select('ST_B10')
  .multiply(0.00341802)
  .add(149.0)
  .subtract(273.15)
  .rename('LST');

/ Calculate NDBI and correlation with LST
ar ndbi = image.normalizedDifference(['SR_B6','SR_B5']).rename('NDBI');
ap.addLayer(ndbi, {min:-0.2, max:0.3, palette:['green','yellow','red']}, 'Built-up Densi
ar combined = ndbi.addBands(lst);
ar correlation = combined.reduceRegion({
  reducer: ee.Reducer.pearsonsCorrelation(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});
rint('NDBI-LST Correlation:', correlation);
```

Figure 1: Code

```
// Visualize LST
Map.addLayer(lst, {min:20, max:45, palette:['blue','yellow','red']}, 'LST');

// Identify Urban Heat Zones (LST > 35°C)
var uhi = lst.gt(35);
Map.addLayer(uhi.updateMask(uhi), {palette:['red']}, 'UHI Zones');

// Mean LST statistics
var stats = lst.reduceRegion({
  reducer: ee.Reducer.mean(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});
print('Mean LST for AOI (°C):', stats);

// UHI zone area
var pixelArea = ee.Image.pixelArea().divide(1e6);
var uhiArea = pixelArea.updateMask(uhi).reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});
print('Urban Heat Zone area (sq km):', uhiArea);

// Export
Export.image.toDrive({
  image: lst,
  description: 'LST_Chennai_Map',
  scale: 30,
  region: aoi
});
```

Figure 2: Code

Results:

The following outputs were generated from the LST-based Urban Heat Island analysis for Chennai district:

1. LST Map

The Land Surface Temperature map (Figure 2) shows the spatial distribution of surface temperature across Chennai district for 2023, with blue indicating cooler surfaces and red indicating high-temperature zones.

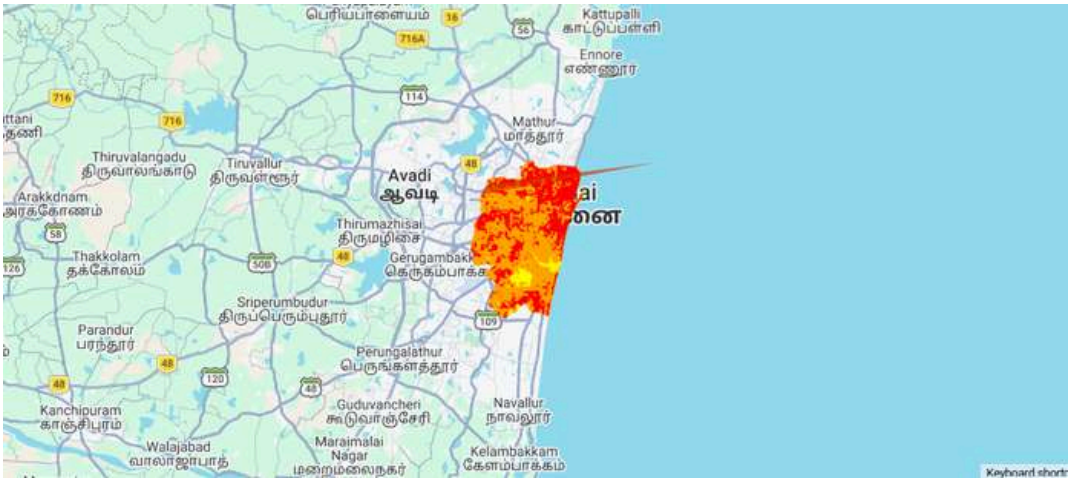


Figure 1: Land Surface Temperature (LST) map of Chennai district, 2023.

2. Urban Heat Zone Map

The Urban Heat Island spatial footprint layer maps localized zones experiencing severe thermal acceleration.

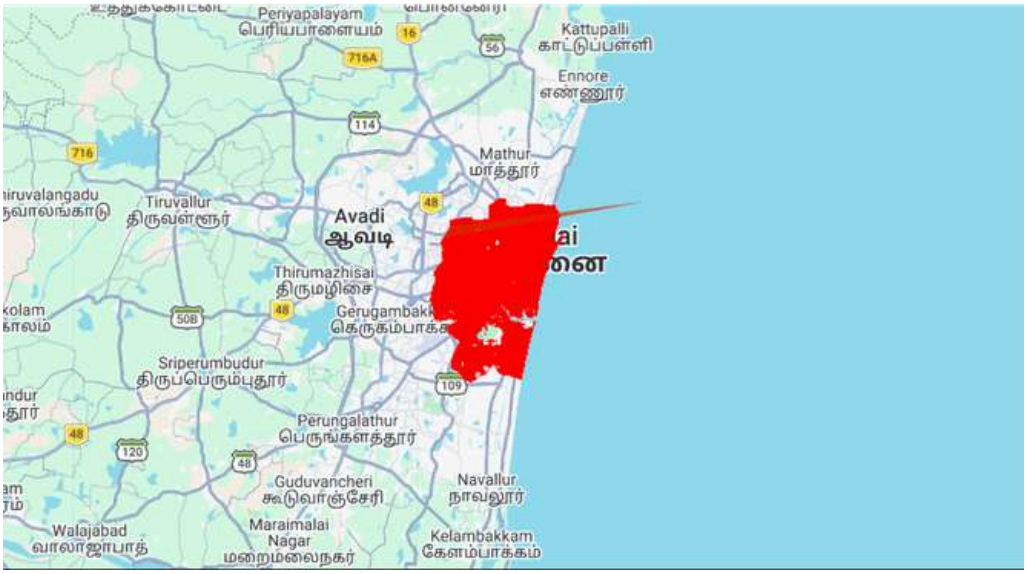


Figure 2: Urban Heat Island (UHI) zones — areas exceeding 35° C LST threshold.

Built-up vs Temperature Maps

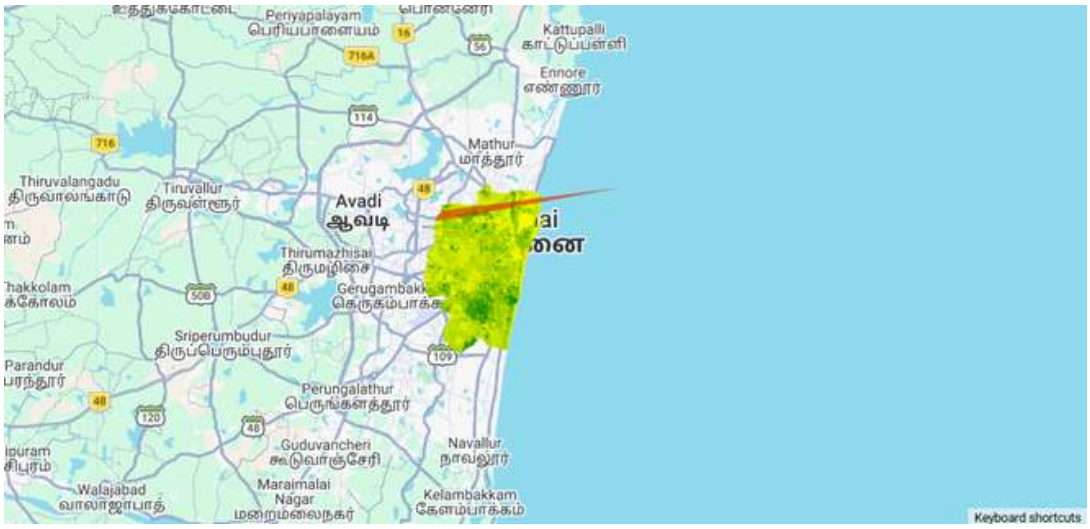


Figure 3: Built up vs Temperature Maps

3. Area and Temperature Statistics

Table 1: LST Metrics and Urban Heat Island Analysis (Chennai, 2023)

Parameter	Value
Total AOI Area	196.63 km^2
Mean LST for AOI	39.34°C
Urban Heat Zone Area (LST > 35°C)	187.05 km^2
Urban Heat Zone Coverage	95.1% of AOI $\frac{187.05}{196.63} \times 100 = 95.1\%$
NDBI Formula	$\frac{SWIR - NIR}{SWIR + NIR}$
NDBI-LST Correlation	0.6470 $= \frac{\sum(NDBI - N\bar{D}BI)(LST - L\bar{S}T)}{\sqrt{\sum(NDBI - N\bar{D}BI)^2 \sum(LST - L\bar{S}T)^2}}$



Figure 3: Console output showing LST statistics for Chennai district.

Conclusion

Urban Heat Island (UHI) analysis using Land Surface Temperature provides valuable insights into temperature variations within urban environments and their relationship with built-up structures. This method enables identification of heat-prone zones, supporting urban planning, climate adaptation, and sustain- able city development.

Although LST estimation may be influenced by atmospheric conditions and surface emissivity assumptions, integrating built-up density and spatial analysis significantly enhances interpretation. This project equips participants with practical skills in thermal remote sensing, urban analysis, and climate-related geospatial applications.

References

- Landsat-8 Collection 2 (LC08/C02/T1 L2) — Google Earth Engine Data Catalog: https://developers.google.com/earth- engine/datasets/catalog/LANDSAT_LC08_C02_T1_L2
- FAO/GAUL 2015 Level-2 Administrative Boundaries — Google Earth Engine Data Catalog: https://developers.google.com/earth- engine/datasets/catalog/FAO_GAUL_2015_level2
- Google Earth Engine Code Editor: https://code.earthengine.google.com
- India Space Academy, Summer Training Program 2026 — Project Work Guidelines and Methodology Document.